

坐标系的微分运动



坐标系的微分运动——微分平移、微分旋转

微分平移：

$$d = d_x i + d_y j + d_z k$$

$${}^T d = {}^T d_x i + {}^T d_y j + {}^T d_z k$$

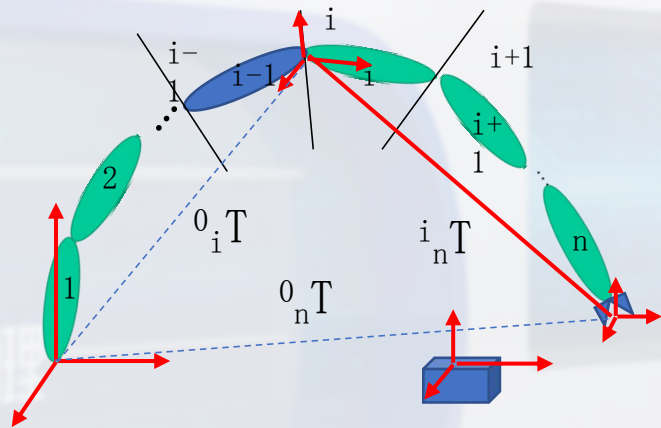
微分旋转：

$$\delta = \delta_x i + \delta_y j + \delta_z k$$

$${}^T \delta = {}^T \delta_x i + {}^T \delta_y j + {}^T \delta_z k$$



坐标系的微分运动——微分平移、微分旋转



$$D = \begin{bmatrix} d_x \\ d_y \\ d_z \\ \delta_x \\ \delta_y \\ \delta_z \end{bmatrix} \quad {}^T D = \begin{bmatrix} {}^T d_x \\ {}^T d_y \\ {}^T d_z \\ {}^T \delta_x \\ {}^T \delta_y \\ {}^T \delta_z \end{bmatrix}$$



相对于基坐标系及当前坐标系的微分运动

相对于基坐标系的微分运动：（坐标系i）

$$dT = (Trans(d_x, d_y, d_z)Rot(f, d\theta) - I)T = \Delta T$$

相对于当前坐标系的微分运动：（坐标系n）

$$dT = T(Trans(d_x, d_y, d_z)Rot(f, d\theta) - I) = T^T \Delta$$



相对于基坐标系及当前坐标系的微分运动

上面两个公式中有形式相同的项：

(Δ 、 ${}^T\Delta$)

(Δ 相对于基坐标系描述——i坐标系)

(${}^T\Delta$ 相对于当前坐标系描述——n坐标系)

$$\Delta = Trans(d_x, d_y, d_z) Rot(f, d\theta) - I$$



绕矢量f (轴) 的通用旋转变换

$$Rot(f, \theta) = \begin{bmatrix} f_x f_x vers \theta + c \theta & f_y f_x vers \theta - f_z s \theta & f_z f_x vers \theta + f_y s \theta & 0 \\ f_x f_y vers \theta + f_z s \theta & f_y f_y vers \theta + c \theta & f_z f_y vers \theta - f_x s \theta & 0 \\ f_x f_z vers \theta - f_y s \theta & f_y f_z vers \theta + f_x s \theta & f_z f_z vers \theta + c \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$vers \theta = 1 - c \theta$$



绕矢量f (轴) 的通用旋转变换

当产生一个微分旋转时 , $c\theta=1$, $\text{vers}\theta=0$, $s\theta=d\theta$

$$\text{Rot}(f, d\theta) = \begin{bmatrix} 1 & -f_z d\theta & f_y d\theta & 0 \\ f_z d\theta & 1 & -f_x d\theta & 0 \\ -f_y d\theta & f_x d\theta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -\delta_z & \delta_y & 0 \\ \delta_z & 1 & -\delta_x & 0 \\ -\delta_y & \delta_x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



微分运动算子

$$\Delta = \text{Trans}(dx, dy, dz) \times \text{Rot}(f, d\theta) - I$$

$$= \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\delta z & \delta y & 0 \\ \delta z & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



$\Delta, {}^T\Delta$ 关系式推导 (1/8)

$$\Delta = \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} S(\delta) & d \\ 0 & 0 \end{bmatrix}$$

$$\Delta T = \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix} T = \begin{bmatrix} S(\delta) & d \\ 0 & 0 \end{bmatrix} \begin{bmatrix} n & o & a & p \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} S(\delta)n & S(\delta)o & S(\delta)a & S(\delta)p + d \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \delta \times n & \delta \times o & \delta \times a & \delta \times p + d \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



$\Delta, {}^T\Delta$ 关系式推导 (2/8)

$$\Delta T = T^T \Delta$$

$$\begin{aligned} {}^T\Delta = T^{-1} \Delta T &= \begin{bmatrix} R^T & \begin{pmatrix} -p \cdot n \\ -p \cdot o \\ -p \cdot a \end{pmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \delta \times n & \delta \times o & \delta \times a & \delta \times p + d \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} R^T(\delta \times n) & R^T(\delta \times o) & R^T(\delta \times a) & R^T[(\delta \times p) + d] \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$



$\Delta, {}^T\Delta$ 关系式推导 (3/8)

$$R^T(\delta \times n) = \begin{bmatrix} n_x & n_y & n_z \\ o_x & o_y & o_z \\ a_x & a_y & a_z \end{bmatrix} (\delta \times n) = \begin{bmatrix} n \cdot (\delta \times n) \\ o \cdot (\delta \times n) \\ a \cdot (\delta \times n) \end{bmatrix} = \begin{bmatrix} 0 \\ \delta \cdot (n \times o) \\ \delta \cdot (n \times a) \end{bmatrix} = \begin{bmatrix} 0 \\ \delta \cdot a \\ -\delta \cdot o \end{bmatrix}$$

混合积

$$a \cdot (b \times c) = b \cdot (c \times a)$$

$$a \cdot (a \times c) = 0$$



$\Delta, {}^T\Delta$ 关系式推导 (4/8)

$$R^T(\delta \times o) = \begin{bmatrix} n_x & n_y & n_z \\ o_x & o_y & o_z \\ a_x & a_y & a_z \end{bmatrix} (\delta \times o) = \begin{bmatrix} n \cdot (\delta \times o) \\ o \cdot (\delta \times o) \\ a \cdot (\delta \times o) \end{bmatrix} = \begin{bmatrix} \delta \cdot (o \times n) \\ 0 \\ \delta \cdot (o \times a) \end{bmatrix} = \begin{bmatrix} -\delta \cdot a \\ 0 \\ \delta \cdot n \end{bmatrix}$$

$$R^T(\delta \times a) = \begin{bmatrix} n_x & n_y & n_z \\ o_x & o_y & o_z \\ a_x & a_y & a_z \end{bmatrix} (\delta \times a) = \begin{bmatrix} n \cdot (\delta \times a) \\ o \cdot (\delta \times a) \\ a \cdot (\delta \times a) \end{bmatrix} = \begin{bmatrix} \delta \cdot (a \times n) \\ \delta \cdot (a \times o) \\ 0 \end{bmatrix} = \begin{bmatrix} \delta \cdot o \\ -\delta \cdot n \\ 0 \end{bmatrix}$$

$$R^T[(\delta \times p) + d] = \begin{bmatrix} n_x & n_y & n_z \\ o_x & o_y & o_z \\ a_x & a_y & a_z \end{bmatrix} [(\delta \times p) + d] = \begin{bmatrix} n \cdot (\delta \times p) + n \cdot d \\ o \cdot (\delta \times p) + o \cdot d \\ a \cdot (\delta \times p) + a \cdot d \end{bmatrix}$$



$\Delta, {}^T\Delta$ 关系式推导 (5/8)

$${}^T\Delta = \begin{bmatrix} 0 & -\delta \cdot a & \delta \cdot o & \delta \cdot (p \times n) + d \cdot n \\ \delta \cdot a & 0 & -\delta \cdot n & \delta \cdot (p \times o) + d \cdot o \\ -\delta \cdot o & \delta \cdot n & 0 & \delta \cdot (p \times a) + d \cdot a \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -{}^T\delta_z & {}^T\delta_y & {}^Td_x \\ {}^T\delta_z & 0 & -{}^T\delta_x & {}^Td_y \\ -{}^T\delta_y & {}^T\delta_x & 0 & {}^Td_z \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$${}^Td_x = n \cdot d + (p \times n) \cdot \delta \quad {}^T\delta_x = n \cdot \delta$$

$${}^Td_y = o \cdot d + (p \times o) \cdot \delta \quad {}^T\delta_y = o \cdot \delta$$

$${}^Td_z = a \cdot d + (p \times a) \cdot \delta \quad {}^T\delta_z = a \cdot \delta$$



$\Delta, {}^T\Delta$ 关系式推导 (5/8)

$$\begin{bmatrix} {}^T d_x \\ {}^T d_y \\ {}^T d_z \\ {}^T \delta_x \\ {}^T \delta_y \\ {}^T \delta_z \end{bmatrix} = \begin{bmatrix} n_x & n_y & n_z & (p \times n)_x & (p \times n)_y & (p \times n)_z \\ o_x & o_y & o_z & (p \times o)_x & (p \times o)_y & (p \times o)_z \\ a_x & a_y & a_z & (p \times a)_x & (p \times a)_y & (p \times a)_z \\ 0 & 0 & 0 & n_x & n_y & n_z \\ 0 & 0 & 0 & o_x & o_y & o_z \\ 0 & 0 & 0 & a_x & a_y & a_z \end{bmatrix} \begin{bmatrix} d_x \\ d_y \\ d_z \\ \delta_x \\ \delta_y \\ \delta_z \end{bmatrix}$$



$\Delta, {}^T\Delta$ 关系式推导 (7/8)

$${}^T\Delta = \begin{bmatrix} 0 & -\delta \cdot a & \delta \cdot o & \delta \cdot (p \times n) + d \cdot n \\ \delta \cdot a & 0 & -\delta \cdot n & \delta \cdot (p \times o) + d \cdot o \\ -\delta \cdot o & \delta \cdot n & 0 & \delta \cdot (p \times a) + d \cdot a \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -{}^T\delta_z & {}^T\delta_y & {}^Td_x \\ {}^T\delta_z & 0 & -{}^T\delta_x & {}^Td_y \\ -{}^T\delta_y & {}^T\delta_x & 0 & {}^Td_z \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$${}^Td_x = n \cdot d + p \cdot (n \times \delta) = n \cdot d + n \cdot (\delta \times p) = n \cdot d - n \cdot (p \times \delta) = n \cdot d - n \cdot S(p)\delta$$

$${}^Td_y = o \cdot d + p \cdot (o \times \delta) = o \cdot d + o \cdot (\delta \times p) = o \cdot d - o \cdot (p \times \delta) = o \cdot d - o \cdot S(p)\delta$$

$${}^Td_z = a \cdot d + p \cdot (a \times \delta) = a \cdot d + a \cdot (\delta \times p) = a \cdot d - a \cdot (p \times \delta) = a \cdot d - a \cdot S(p)\delta$$



$\Delta, {}^T\Delta$ 关系式推导 (8/8)

$$\begin{bmatrix} {}^T d \\ {}^T \delta \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} n \\ o \\ a \end{pmatrix} - \begin{pmatrix} n \\ o \\ a \end{pmatrix} S(p) \\ 0 \quad \begin{pmatrix} n \\ o \\ a \end{pmatrix} \end{bmatrix} \begin{bmatrix} d \\ \delta \end{bmatrix}$$

$${}^T \delta_x = \delta \cdot n = n \cdot \delta$$

$${}^T \delta_y = \delta \cdot o = o \cdot \delta$$

$${}^T \delta_z = \delta \cdot a = a \cdot \delta$$

$$\begin{bmatrix} {}^T d \\ {}^T \delta \end{bmatrix} = \begin{bmatrix} R^T & -R^T S(p) \\ 0 & R^T \end{bmatrix} \begin{bmatrix} d \\ \delta \end{bmatrix}$$



微分运动变换关系 (小结1/2)

$$dT = (Trans(d_x, d_y, d_z)Rot(f, d\theta) - I)T = \Delta T$$

$$dT = T(Trans(d_x, d_y, d_z)Rot(f, d\theta) - I) = T^T \Delta$$

$${}^T \delta x = \delta \cdot n$$

$${}^T \delta y = \delta \cdot o$$

$${}^T \delta z = \delta \cdot a$$

$${}^T dx = n \cdot [(\delta \times p) + d]$$

$${}^T dy = o \cdot [(\delta \times p) + d]$$

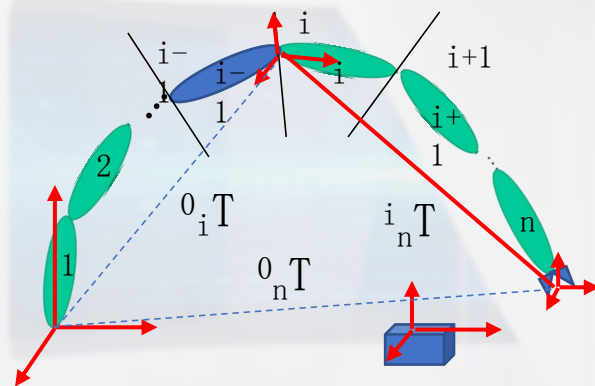
$${}^T dz = a \cdot [(\delta \times p) + d]$$

$$T = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



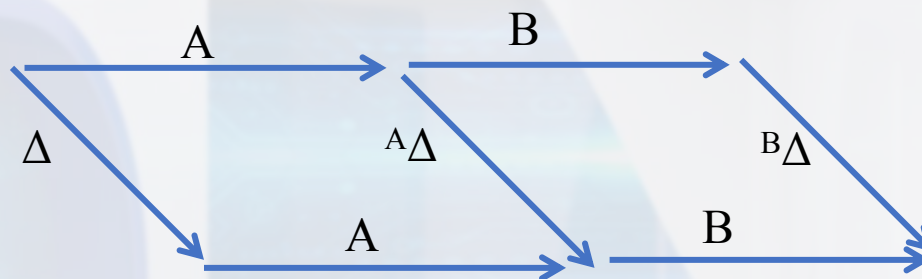
微分运动变换关系（小结1/3）

$$\Delta T = T^T \Delta$$



微分运动变换关系（小结2/3）——变换关系图

$$\Delta AB = AB^B \Delta$$
$$A^A \Delta B = AB^B \Delta$$



微分运动变换关系（小结3/3）——变换关系图

$${}^A\Delta = \Delta A$$

$$T^T\Delta = \Delta T$$

$$B^B\Delta = {}^A\Delta B$$

$${}^A\Delta = \begin{bmatrix} A^T & -A^T S(p) \\ 0 & A^T \end{bmatrix} \Delta$$

$$B\Delta = \begin{bmatrix} B^T & -B^T S(p) \\ 0 & B^T \end{bmatrix} {}^A\Delta$$



END

